

A Four cooperation dilemmas

Extractive dilemma

depletion



Overuse of a shared resource depletes it.

In humans

overfishing, overgrazing

In LLM agents

API budgets, shared compute, tool calls, context windows

Contributive dilemma

no upkeep



Beneficiaries fail to maintain community goods.

In humans

open-source upkeep, unpaid moderation

In LLM agents

helping at a personal cost, cleanup, verifying, volunteering

Second-order dilemma

who watches whom?



Monitoring is itself costly to provide.

In humans

peer monitoring, costly security, punishment

In LLM agents

auditing, sanctioning agents, setting new norms

Repair dilemma

restoration



Who bears the cost of fixing harm?

In humans

restitution, policy reform, disaster recovery

In LLM agents

cleaning corrupted memory, retractions, undoing changes, restoring what was lost

B Design space of agent institutions

